



Bagging

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Working in groups is better
- Bagging
- Bootstrapping
 - With replacement
 - Without replacement

Place your
Webcam Video here
Size 100%

Working in a group is better



- Manage high variance by having an **ensemble** of classifiers that will vote together
 - If everyone votes we won't go radically into any direction
 - Everyone voting together *smooths* our decision function



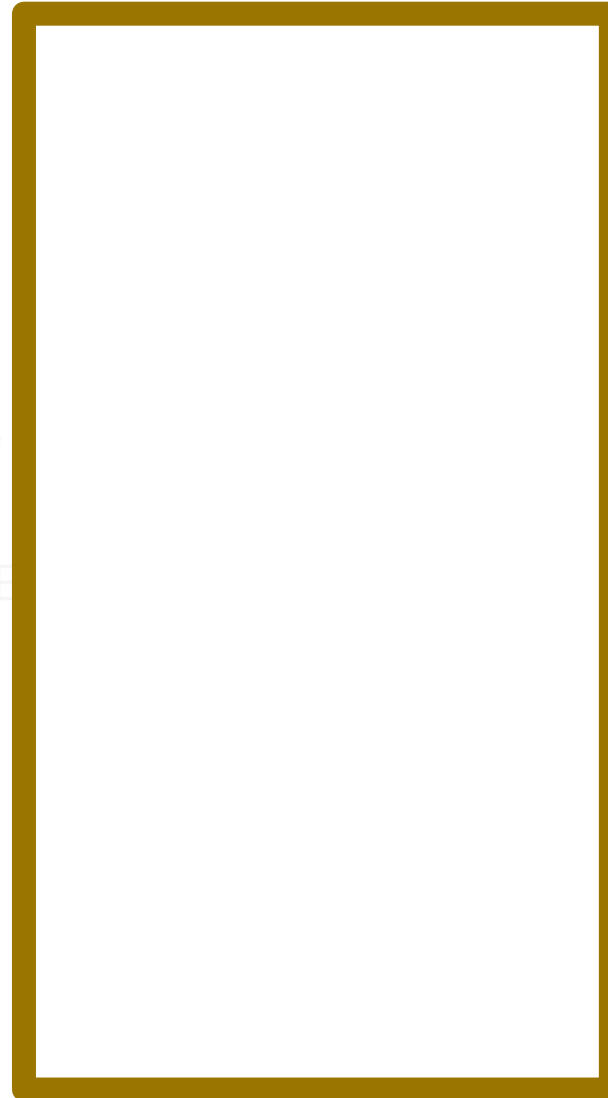
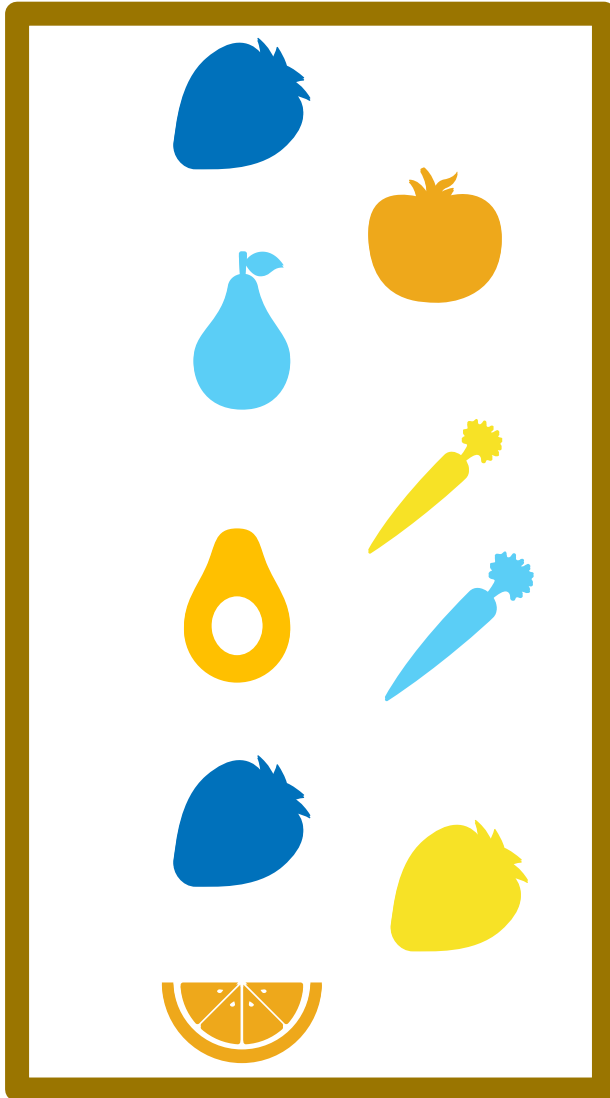
Ensemble Strategy

- What characteristics do we want the individual learners to have? The entire ensemble?
 - Each individual should have a different opinion
 - The ensemble should produce a good answer
- How do we make different opinions?

Bagging – Bootstrap Sampling

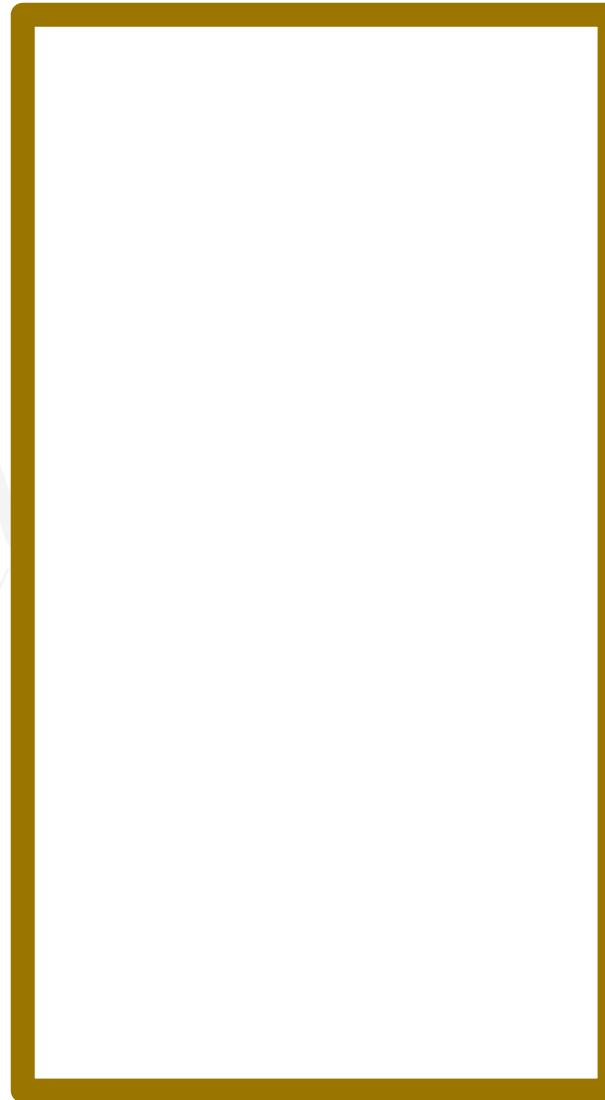
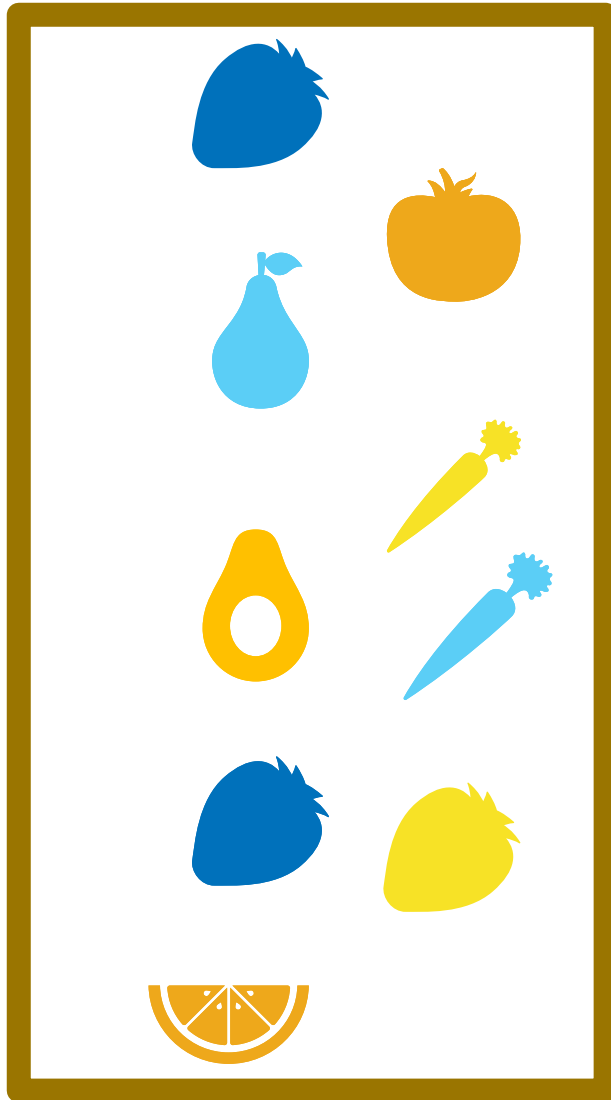
- Randomly “bootstrap” resample things to draw new distributions (make bags!)
- Original set: [apple_1, orange, melon, watermelon, apple_2, pineapple]
 - Make subsets:
 - Bag1: [apple_2, orange, melon, apple_1]
 - Bag2: [orange, melon, watermelon, apple_2]

Without Replacement



- We finish with a new dataset that is a subset of the original dataset

With Replacement



- We finish with a new dataset that is the same size, but a variation of the original data

Conclusion

- Ensembles should have many opinions
- Bagging helps diversify opinions
- Bootstrap sampling
 - w/ replacement
 - w/o replacement

Place your
Webcam Video here
Size 100%



Bagging: Random Forests

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- What is a random forest
- Bootstrapping features
- Ensembles produce single answers with voting
- Bagging is a meta algorithm
- Pros and Cons

Place your
Webcam Video here
Size 100%

Bagging + Trees = Random Forests

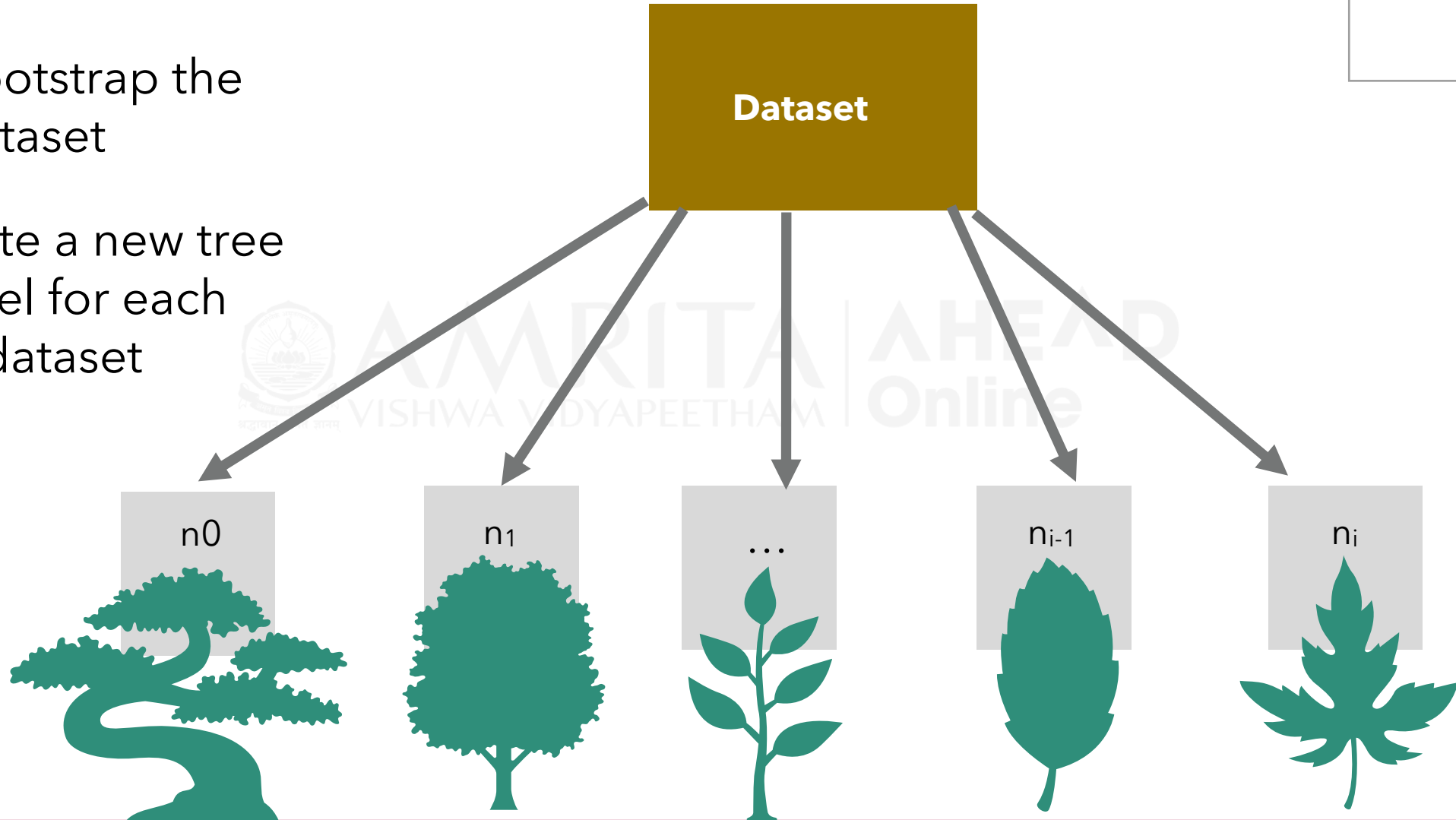
- Sample n-datasets with bootstrapping
- Can bootstrap with/without replacement
- For each dataset build a decision tree



Forests of Trees

Bootstrap the
dataset

Create a new tree
model for each
nth-dataset



Bootstrap Features



Size	# Bedrooms	Color	School District Rating
10	3	Red and Brown	Good
20	2	White	Bad
18	3	Black and Blue	Very Bad
12	6	Green	Very Good

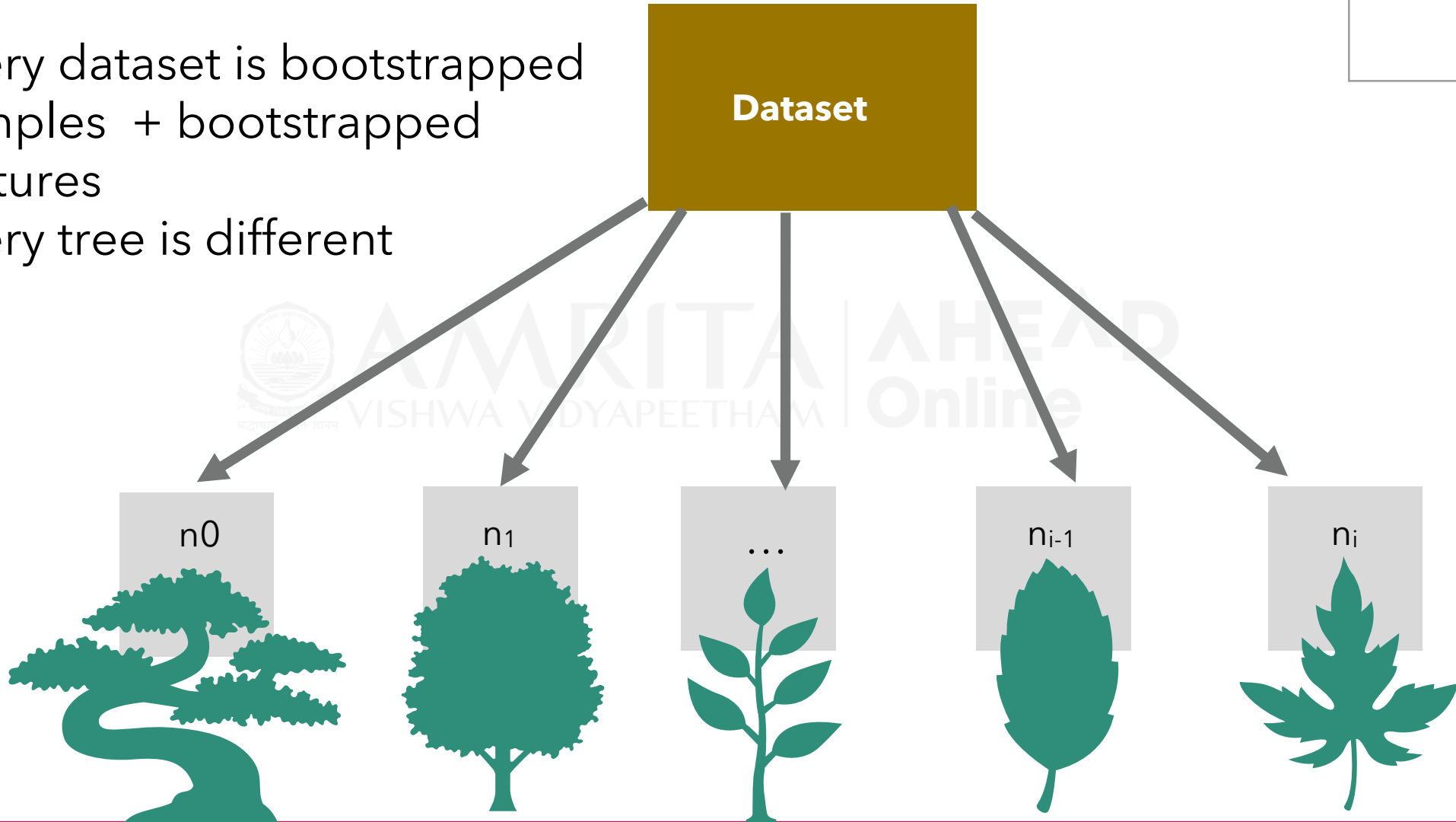
Drop Columns Randomly



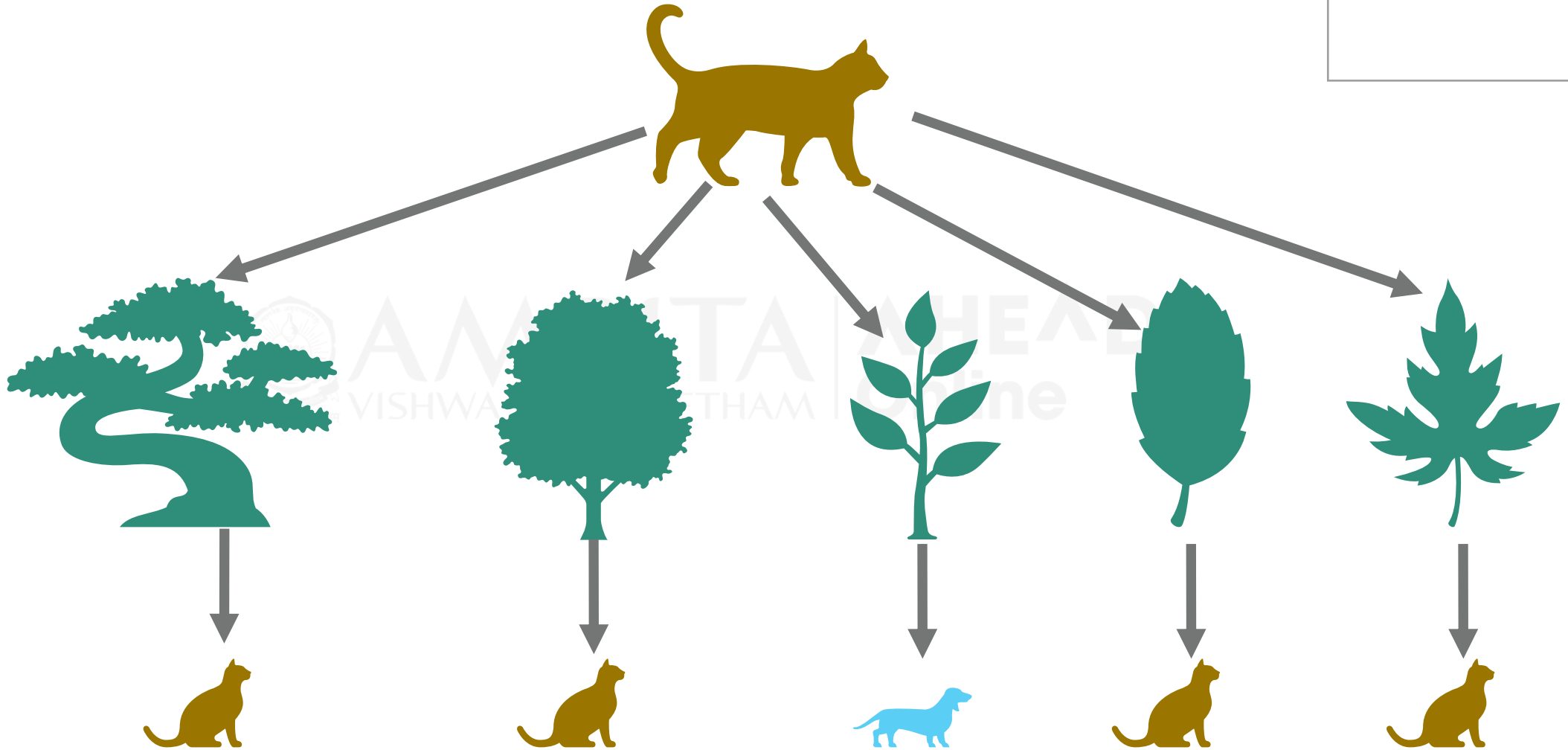
Size	# Bedrooms	Color	School District Rating
10	3	Red and Brown	Good
20	2	White	Bad
18	3	Black and Blue	Very Bad
12	6	Green	Very Good

Forests of Trees

- Every dataset is bootstrapped samples + bootstrapped features
- Every tree is different



Vote to make a prediction



Random Forest Hyper-Parameters

- All the decision tree options PLUS
- N-estimators : How many trees?
- Max_features: The number of features to consider when looking for the best split:
- Replacement : whether or not to allow replacement.

Bagging can be done on any model

- It's most commonly done with decision trees
 - They are easy to make randomly and plentiful
- You can do it with any model though, just resample and you are set
- Bagging is a "meta" algorithm

Pro's and Con's

- Random forests are groups of decision trees built on bootstrapped datasets.
- Pro's:
 - Having many estimators smooths the model, making it less likely to overfit
 - Get better performance overall
- Cons
 - It's computationally more expensive

Conclusion

- Looked at the meta algorithm of bagging
- How it works on Random Forests
- Overall a great idea and works very well
- Can be expensive

Place your
Webcam Video here
Size 100%











Boosting

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Can we do an ensemble method that gets smarter as it goes?
- Adaboost

Place your
Webcam Video here
Size 100%

Boosting (AdaBoost)

- Train a series of models that slowly get smarter.
- Each model will learn from the mistakes of the previous model (gets boosted!)
- The final model is a combination of all these models

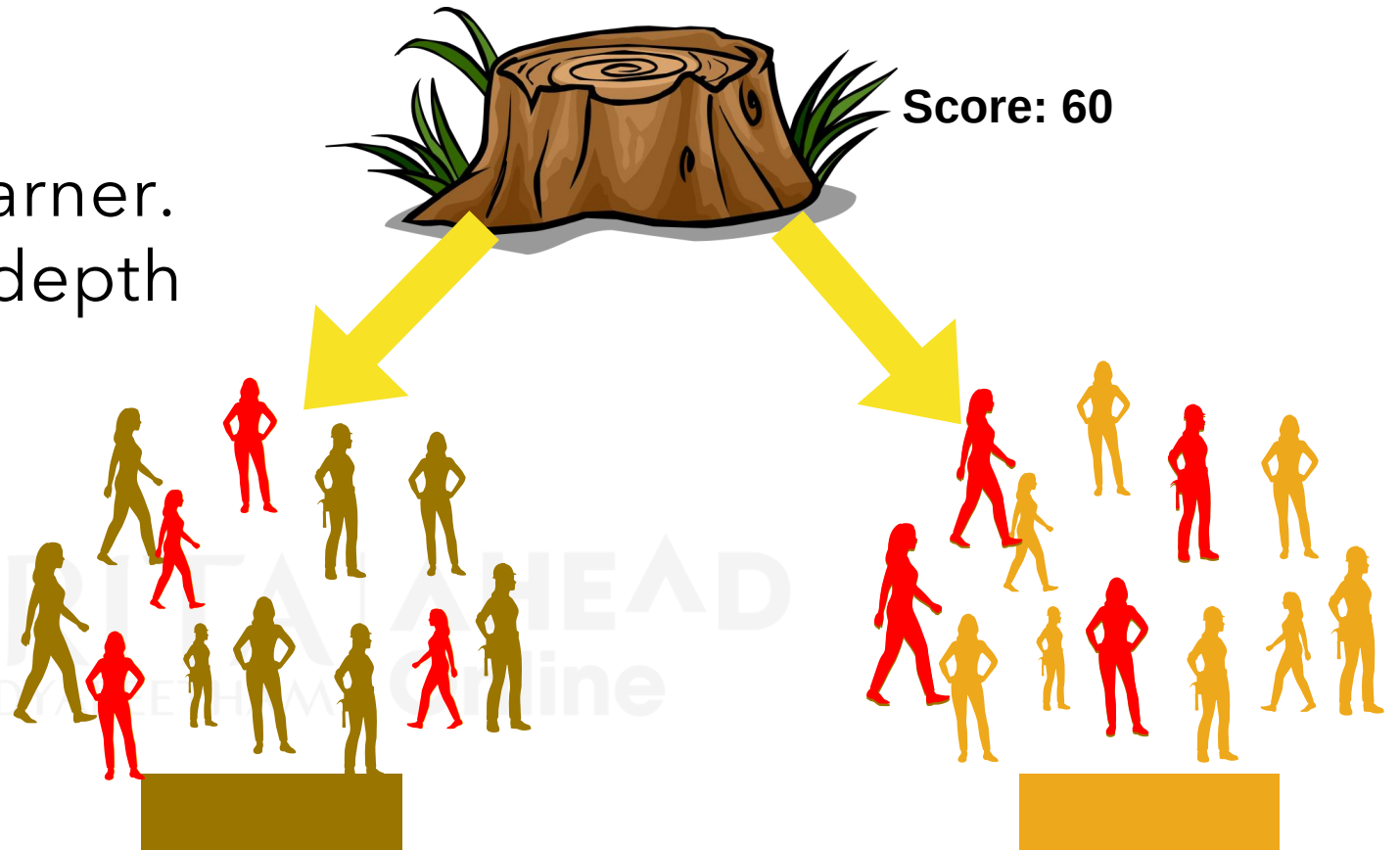


Boosting (Adaboost)

Step 1: Make very weak learner.
Decision Tree Stump (max-depth is 1)

Step 2: Score the stump
- how good was it?
Weight the errors that
the stump made

Step 3 Resample the dataset



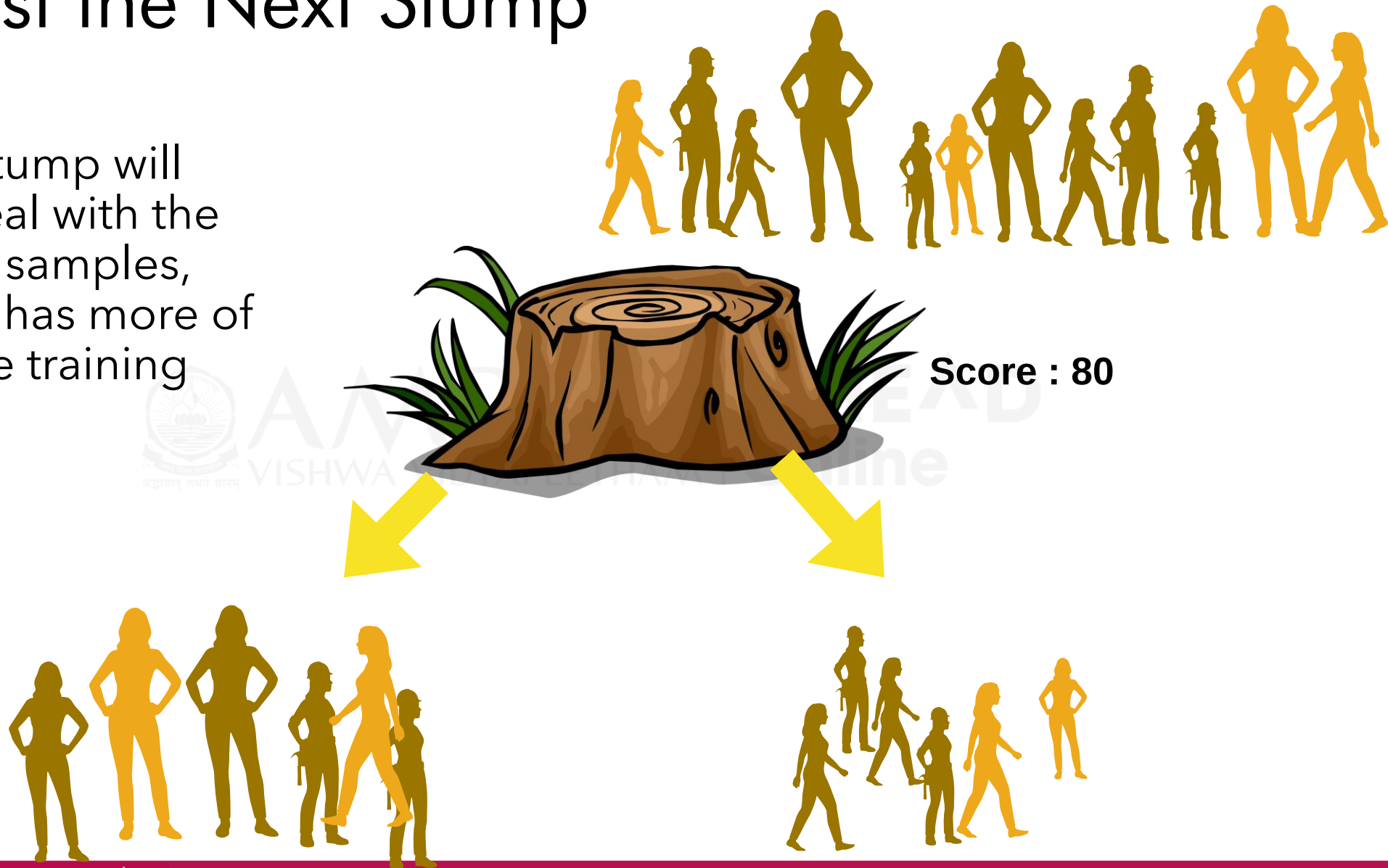
Resample



- Weight the errors, and during resampling give them a higher probability of being chosen.

Boost the Next Stump

The next stump will learn to deal with the "problem" samples, because it has more of them in the training data



Combine all Stumps



Score : 56



Score : 23



Score : 50



Score : 80

AMRITA | AHEAD
VISHWA VIDYAPEETHAM Online

- Make predictions by combining all stumps and weighting their predictions by their score

Adaboost Properties



- Successive stumps
- Each one is affected by the previous one
 - Each dataset has more of the “hard problems” (samples we missed)
 - This increases the models ability to learn those problems.
- Tends to do very well – but can overfit

Gradient Boosting

- Similar to adaboost, but uses the mathematical principle of gradients to build many trees that “step” in the correct direction.
- XGBoost and LightGBM are the two most popular gradient boosting algorithms.
 - Very popular – state of the art in structured machine learning problems.

Pro's and Con's

- Pro's
 - Better Performance
- Cons:
 - More things to manage
 - Computationally expensive
 - Possibility of overfitting



Conclusion

- Looked at the basics of adaboost
- Key idea: using resampling to force the model to learn 'harder' problems
- Works well in practice
- Can overfit

Place your
Webcam Video here
Size 100%